Tutorial-2.

A mobile service provider wants to classify sms message as spam or Not spam using a simple neutral network (perceptron)

x_1 = specific words and phrases

x_2 = Links and Attachments

x_3 = Bias -> constant
contral the boundary of models

x_1	x_2	x_3	Target(y) (yes/no) question
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

① Inputs : x_1 & x_2.

② weights : w_1 & w_2.

③ Bias (b)

④ weighted sum : $z = x_1 w_1 + x_2 w_2 + b$.

$= \sum_{i=1}^{z} x_i w_i + b$

⑤ Activation function (Here : step function)

$$\hat{y} = \begin{cases} 0; & z < 0 \\ 1; & z \geq 0 \end{cases}$$

⑥ Updated weights

$w_i^{new} = w_i^{old} + \eta (y - \hat{y}) x_i$

η = Learning rate = 1

y = actual target / True Target

$\hat{y}$ = predicted output / Target

$x_1 = 4, \quad x_2 = 2, \quad b = 1$

Let $w_1 = 0, \quad w_2 = 0 \quad \eta = 1$.

Inputs =

$z = ?$

$z = w_1 x_1 + w_2 x_2 + b$

$z = 0(-1) + (0)(2) + 1$

$\boxed{z = 1}$ so check $\boxed{y^n = 1}$

The y^n is matched with target (y) in 1st row

Iteration = epoch -> 10

③ An email service provider wants to classify incoming emails as spam or not spam using a single network model.

The training dataset is shown below

x_1	x_2	x_3	Target (Y)
5	3	1	0
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

① Inputs

$x_1 = 5, \; x_2 = 3, \; b = 1$

Let $w_1 = 0, \; w_2 = 0, \; \eta = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$z = (0)(5) + (0)(3) + 1$

$z = 1$ so $\boxed{y^n = 1}$

$w_1 new = w_1 old + \eta (y - y^n) x_1$

$w_1 new = (0) + 1 (0-1) 5$

$= 0 + (-5)$

$w_1 new = -5$

$w_2 new = w_2 old + \eta (y - y^n) x_2$

$= 0 + (0-1) 3$

$w_2 new = -3$

$w_1 = -5,\ w_2 = -3,\ \eta = 1,\ x_1 = 1,\ x_2 = 0,\ b = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= (-5)(1) + (-3)(0) + 1$

$= -5 + 1$

$\boxed{z = -4}$ $\boxed{y^{\wedge} = 0}$ -> The output of old is predicted.

$w_{new} = (-5) + 1(y - y^{\wedge})x_1$

$= (-5) + 1(0 - 0)1$

$\boxed{w_{1new} = -5}$

If we update weights after output is correct then the same weights will comes.

An e-commerce company predicts whether a customer will make a purchase based on time spent on website (minutes) and number of items viewed.

Dataset :-

Time on site (x_1)	items viewed. (x_2)	Bios (x_3)	Target (purchase = 1, No = 0)
2	3	1	0
15	25	1	1
8	12	1	1
3	5	1	0
12	20	1	1

① Inputs :-

$x_1 = 2,\ x_2 = 3,\ b = 1,\ w_1 = 0,\ w_2 = 0,\ \eta = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= (0)(2) + (0)(3) + 1$

$\boxed{y^{\wedge} = 1}$ $z = 1$

$w_1 new = w_1 old + \eta (y - \hat{y}) x_1$

$w_1 = 0 + 1 (0-1) 2.$

$\boxed{w_1 = -2}$

$w_2 new = 0 + 1 (0-1) 3$

$w_2 = -3$

2nd case

$w_1 = -2, w_2 = -3, b = 1, x_1 = 15, x_2 = 25, \eta = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= (-2)(15) + (-3)(25) + 1$

$= -30 + -75 + 1$

$= -104$

$\boxed{\hat{y} = 0}$

$w_1 new = w_1 old + \eta (y - \hat{y}) x_1$

$w_1 new = -2 + 1 (1-0) 15$

$= -2 + 15$

$\boxed{w_1 = 13}$

$w_2 new = w_2 old + \eta (y - \hat{y}) x_2$

$= -3 + 1 (1-0) 25$

$= -3 + 25$

$\boxed{w_2 = 22}$

$w_1 = 13, w_2 = 22, \eta = 1$

$x_1 = 8, x_2 = 12, b = 1$

$z = w_1 x_1 + w_2 x_2 + b.$

$z = (13)(8) + (22)(12) + 1$

$z = 104 + 246 + 1$

$\boxed{z = 351}$ $\boxed{\hat{y} = 1}$

At the third row we have predicted.

An email service provider wants to classify incoming emails as spam or not spam using a single layer network model with 5 hidden neutrons.

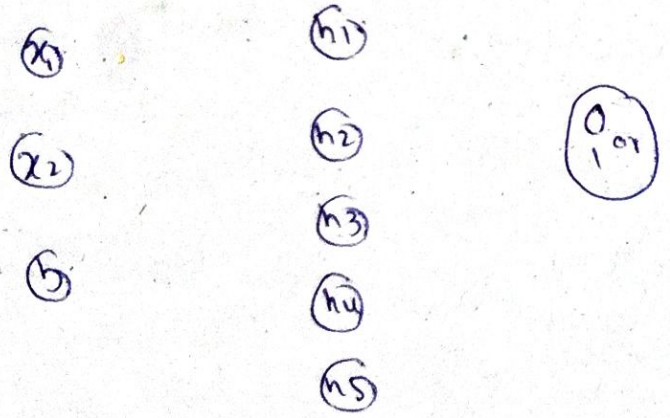
The training dataset is shown below.

$b = w_0 = x_3$

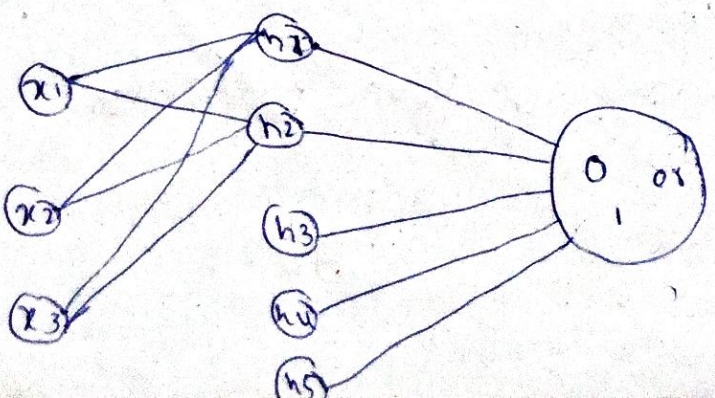
x_1	x_2	x_3 (b)	Target (Y)
5	3	1	1
1	0	1	0
7	5	1	1
2	1	1	0
6	4	1	1

Hidden layers are which is located like a mediator b/w Inputs and outputs.

Single layer neural network.



everything is works / performed by hidden neuron each hidden neuron is connected to all the Inputs (x_1, x_2, b) in a single input layer.



Ex:

	w_1	w_2	b
h_1	0	0	1
h_2	0.1	-0.2	1
h_3	0.3	0.1	1
h_4	0.2	-0.2	1
h_5	0.4	0.1	1

h_1

$$z_1 = w_1 x_1 + w_2 x_2 + b$$
$$= (0) + (0) + 1$$
$$= 1$$

$h_2 \Rightarrow$

$$z_2 = w_1 x_1 + w_2 x_2 + b$$
$$z = (0.1)(5) + (-0.2)(3) + 1$$
$$= 0.9$$

$h_3 \Rightarrow$

$$z_4 = (0.3)(5) + (0.1)(3) + 1$$
$$= 2.8$$

$h_4 \Rightarrow$

$$z_4 = (0.2)(5) + (-0.2)(3) + 1$$
$$= 1.4$$

$h_5 \Rightarrow$

$$z_5 = (0.4)(5) + (0.1)(3) + 1$$
$$= 3.3$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{(-z)}} = \frac{1}{1+e^{(-1)}} = 0.74$$

$$h_2 = \sigma(z_2) = \frac{1}{1+e^{(-z)}} = \frac{1}{1+e^{(-0.9)}} = 0.72$$

$$h_3 = \sigma(z_3) = \frac{1}{1+e^{(-z)}} = \frac{1}{1+e^{(-2.8)}} = 0.95$$

$h_4 = \sigma(z_4) = \frac{1}{1+e^{(-z)}} = \frac{1}{1+e^{(-1.4)}} = 0.80$

$h_5 = \sigma(z_5) = \frac{1}{1+e^{(-z)}} = \frac{1}{1+e^{(-3.3)}} = 0.97$

$z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$

$\sum_{i=1}^{N} = h_i v_i$

$w \rightarrow$ hidden layers and Inputs

$v \rightarrow$ hidden layers and output

$v = 0.1, 0.2, 0.3, 0.4, 0.5, -0.1, -0.5$ any val.

$\sigma(z_{out}) = ?$

$(z_{out}) = (0.74)(0.1) + (0.72)(0.2) + (0.95)(0.3) + (0.80)(-0.2) + 0.97(0.5) = 0$

$\sigma(z_{out}) = \frac{1}{1+e^{(-828)}}$

$= 0.828$

$\sigma(z_{out}) \Rightarrow 0.999 \simeq 1.0$

$\sigma(z_{out}) > 0.5$ $\boxed{y^n = 1}$

if $z(out) \geq 0.5 \Rightarrow 1$
$< 0.5 \Rightarrow 0$ } mission will detects.

$w_i new = w_i old + \eta (y - y^n) x_i$

$v_i new = v_i old + \eta (y - y^n) h_i$

If the output is less than 0.5 then we want to update the w, v values both/ any one of it

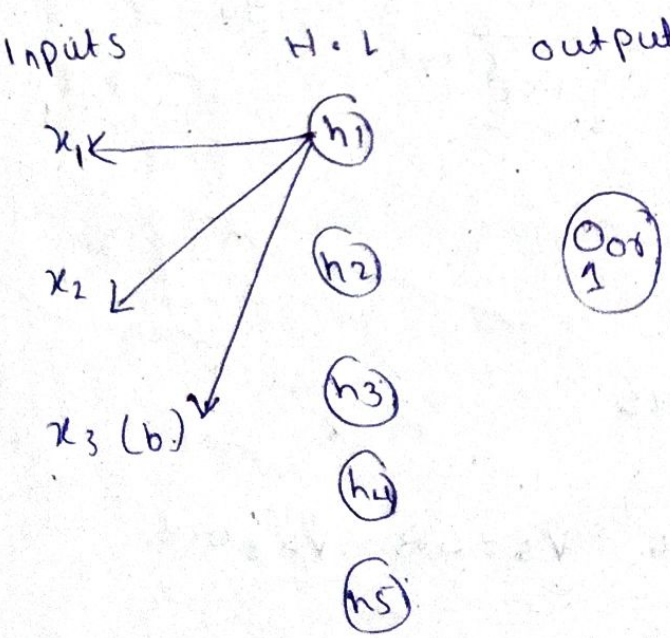
A mobile service provider wants to classify sms message as spam or Not spam using a single-layer neural network with 5 hidden neurons in the layer.

x_1 = specific words and phrases

x_2 = links and attachments

x_3 = bias.

x_1	x_2	x_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0



h1

$z_1 = w_1x_1 + w_2x_2 + b$

$z = (0.1)(4) + (0.2)(2) + 1$

$= 1.8$

$\underline{h2}$

$z_2 = w_1x_1 + w_2x_2 + b$

$= (-0.2)(4) + (0.3)(2) + 1$

$= 0.8$

$\underline{h3}$

$z_3 = w_1x_1 + w_2x_2 + b$

$= (0.4)(4) + (-0.1)(2) + 1$

$= 2.4$

$\underline{h4}$

$z_4 = w_1x_1 + w_2x_2 + b$

$= (0.3)(4) + (0.2)(2) + 1$

$= 2.6$

$\underline{h5}$

$z_5 = w_1x_1 + w_2x_2 + b$

$= (0.2)4 + (-0.4)(2) + 1$

$= 1$

$h_1 = \sigma(z_1) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{(-1.8)}} = 0.86$

$h_2 = \sigma(z_2) = 0.69$

$h_3 = \sigma(z_3) = 0.92$

$h_4 = \sigma(z_4) = 0.93$

$h_5 = \sigma(z_5) = 0.73$

$v = 0.1, \quad v_2 = -0.2 \quad v_3 = 0.3 \quad v_4 = 0.4$

$v_5 = -0.3$

$z_{out} = h_1v_1 + h_2v_2 + h_3v_3 + h_4v_4 + h_5v_5$

$= 0.86(0.1) + 0.69(-0.2) + (0.92)(0.3)$

$+ 0.93(0.4) + 0.73(-0.3)$

$= 0.38$

$$\sigma(z_{out}) = \frac{1}{1+e^{-(0.38)}}$$

$$a(z_{out}) = 0.59 \sim 0.60$$

$$\boxed{\begin{array}{l} a < 0.5 \rightarrow \hat{y} = 0 \\ a \geq 0.5 \rightarrow \hat{y} = 1 \end{array}} \Rightarrow a(z_{out}) \geq 0.5$$

so $\boxed{\hat{y} = 1}$

$\hat{y}$ or y predicted.

$$\boxed{v_{i\,new} = v_{i\,old} + \underbrace{\eta}_{\text{Learning Rate}} \underbrace{(y}_{\text{True}} - \underbrace{\hat{y})}_{\text{predicted}} h_i}$$

$$\boxed{w_{i\,new} = w_{i\,old} + \eta (y - \hat{y}) x_i}$$

we can update both v, w (or) any one of it, if output not predicted,